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JupyterHub Training Platform on Kubernetes

Chris Green – STFC Scientific Computing Department

Overview

- What is JupyterHub?
- Why Kubernetes?
- How do we deploy it?



What is JupyterHub?

- Provides pre-configured Jupyter notebooks
- Accessible by a group of users
 - Students
 - Researchers
 - Data Scientists
- Users get access to compute



Server Options

- **Default environment**

This is a simple default environment

Start



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Simple

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Launcher

Notebook

Python 3 (ipykernel)

Console

Python 3 (ipykernel)

Other

Launcher 1

UKRI

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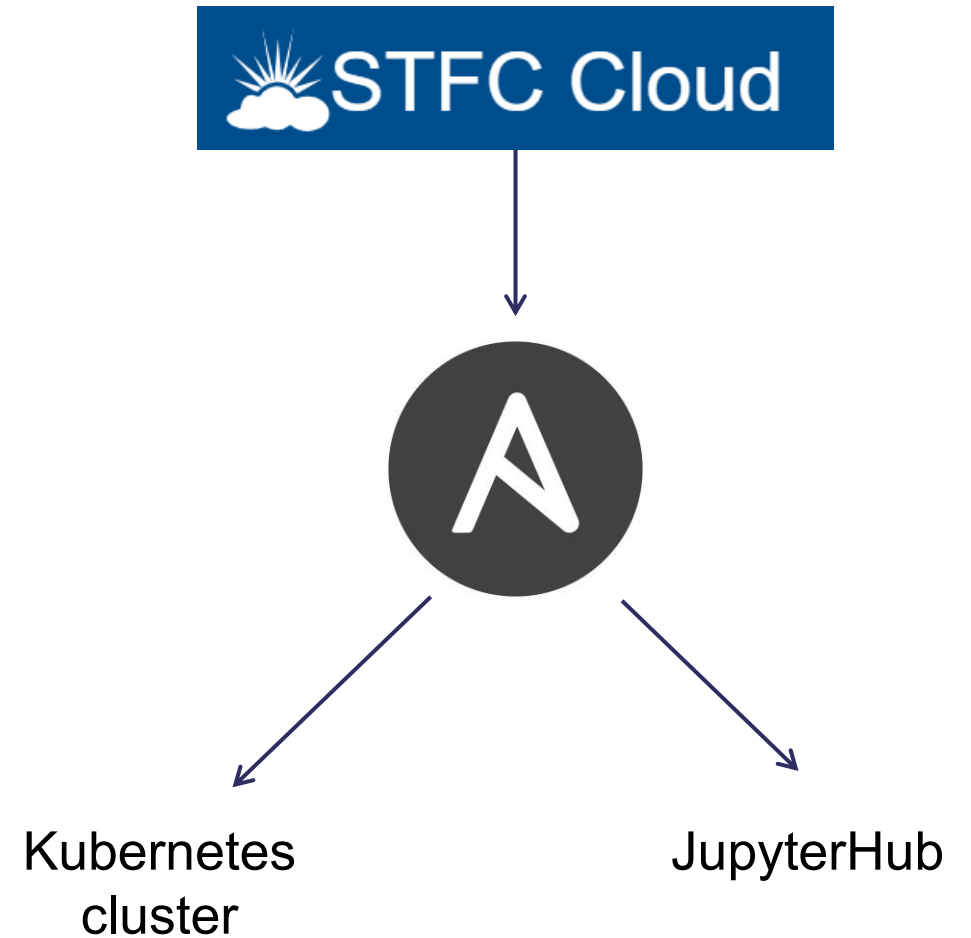
Why Kubernetes?

- Easy to deploy on multiple node
- Different environments get different hardware
 - GPU, CPU, RAM
- Multiple users per node
- Multiple services per node
 - Isolated environments
- Automatic re-deployment upon software crashes
- Handles networking for you
- Monitoring & Alerting



How do we deploy it?

- STFC Cloud
 - Multiple node sizes available
 - CPU, GPU
- Ansible
 - Configuration management
 - Configure Kubernetes deployment
- Deploy Kubernetes cluster
- Deploy JupyterHub onto Kubernetes cluster
 - Longhorn storage



```
management_cluster_name: "cg-management-cluster"
```

```
management_cluster:
```

```
  floating_ip: "123.456.78.90"
```

```
  control_plane_flavor: "l3.nano"
```

```
  control_plane_count: 3
```

```
  node_groups:
```

```
    - name: "jhub-cluster"
```

```
      size: "2"
```

```
      flavor: "l3.micro"
```

```
admin_password:
```

```
number_of_users:
```

```
username:
```

```
user_password:
```

```
profiles:
```

```
  - profile1:
```

```
    display_name: "Default environment"
```

```
    description: "This is a simple default environment"
```

```
    default: true
```

```
    image: jupyter/minimal-notebook
```

```
    node_groups: []
```

```
    gpu_required: false
```

```
    gpu_display_drivers: false
```

```
    commands:
```

```
      -
```

```
      -
```



Summary

- What is JupyterHub?
 - Training platform for multiple users
- Why Kubernetes?
 - Easily handle multiple nodes
 - Re-deploy on errors/crashes
 - Isolate environments
- How do we deploy it?
 - Ansible
 - Easy deployment and configuration





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